Project Report

Gruha Alankara: AI-Powered Interior Design Platform with AR Integration

Track- Agentic AI Application Devloper with IBM

Submitted By

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1. Introduction

Interior design plays an important role in enhancing the aesthetics and functionality of living spaces. However, professional interior design services are often expensive and inaccessible for many individuals. With the advancement of Artificial Intelligence and Augmented Reality technologies, it is now possible to provide intelligent design recommendations automatically.

The **Gruha Alankara** platform is an AI-driven interior design system that allows users to analyze their room, select preferred design styles, and receive personalized design recommendations. The system integrates computer vision techniques, AI reasoning modules, and modern web technologies to provide an interactive design experience.

Users can upload room images or capture them using a live camera feed, after which the system analyzes the room layout and generates furniture placement

suggestions, color palettes, and layout recommendations. The platform also provides a furniture catalog and booking functionality to help users implement the recommended designs.

The project demonstrates how modern technologies such as **Flask, AI models, and AR-based visualization** can transform the interior design process.

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2. Objectives

The main objectives of the Gruha Alankara system are:

- To develop an AI-powered platform that generates interior design suggestions automatically.
- To provide users with a user-friendly interface for uploading and analyzing room images.
- To integrate AI modules that analyze room characteristics and recommend furniture placement.
- To allow users to explore furniture catalogs and design options based on their preferences.
- To demonstrate the integration of computer vision, AI reasoning, and web technologies in a practical application.

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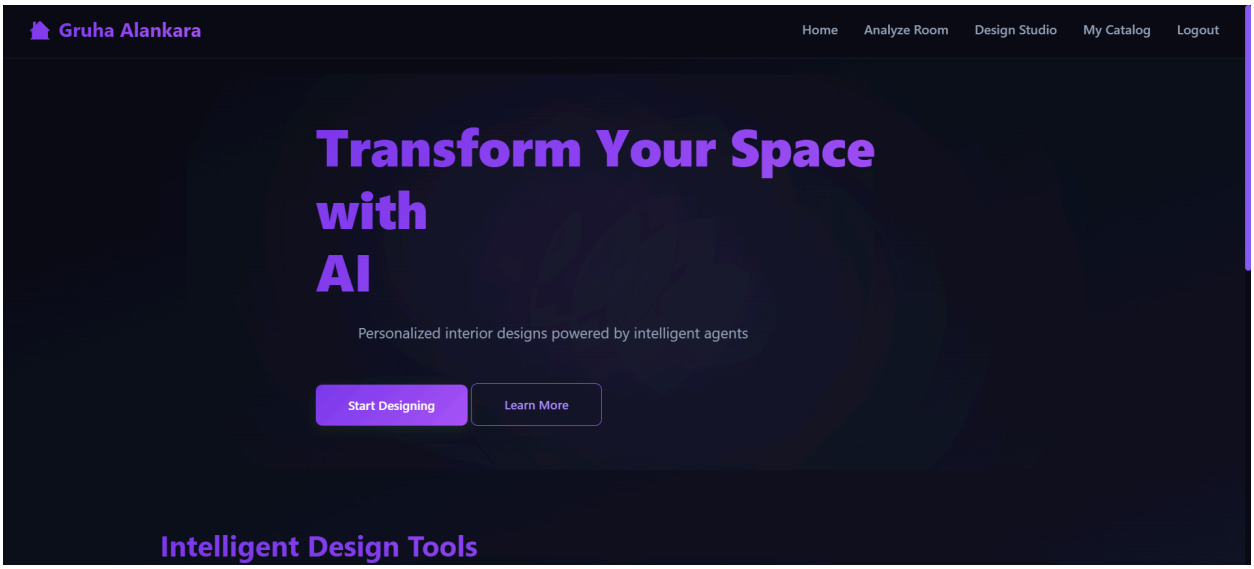
3. System Overview

Gruha Alankara is designed as a **web-based application** that combines frontend interfaces with backend AI processing modules.



The system works in the following stages:

1. User authentication and account creation.
2. Upload or capture a room image.
3. Analyze the room using AI modules.
4. Generate design recommendations.
5. Display furniture suggestions and layout options.
6. Allow users to explore furniture catalogs and booking options.

The platform uses a modular architecture where each AI component performs a specific task such as room analysis, style recommendation, or furniture optimization.

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4. Technology Stack

The project uses the following technologies:

Programming Language

Python

Backend Framework

Flask

Database

SQLite with SQLAlchemy ORM

Frontend

HTML, CSS, JavaScript

AI and Machine Learning

Transformer models and custom AI modules

Computer Vision

Image processing techniques for room analysis

APIs

MediaDevices API for camera access

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5. System Architecture

The application follows a modular architecture consisting of multiple components:

1. User Interface Layer

Provides interactive web pages where users can upload images, select styles, and view design results.

2. Backend Application Layer

Handles requests, processes user inputs, interacts with the database, and coordinates AI modules.

3. AI Processing Layer

Includes specialized modules responsible for analyzing rooms and generating design recommendations.

4. Data Storage Layer

Stores user data, design history, and furniture catalog information in a relational database.

This layered architecture ensures scalability and maintainability of the system.

(Optional diagram can be added here but not necessary)

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6. Database Design

The system uses a relational database to store user information, design data, and booking records.

Main Tables

Users

- User ID
- Username
- Email
- Password hash
- Account creation timestamp

Designs

- Design ID
- User ID
- Uploaded image path
- Selected style theme
- AI-generated design output
- Creation timestamp

Furniture

- Furniture ID
- Item name
- Category

- Price
- Image URL

Bookings

- Booking ID
- User ID
- Furniture ID
- Booking status
- Booking date

These tables allow the system to track user activity, generated designs, and furniture purchases.

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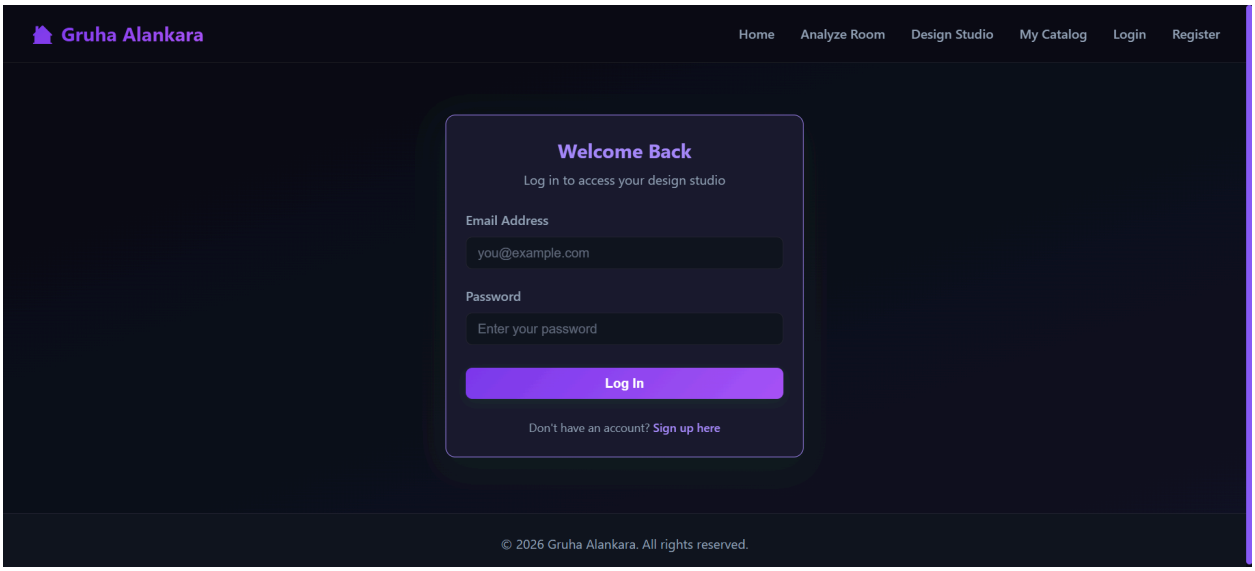
7. System Modules

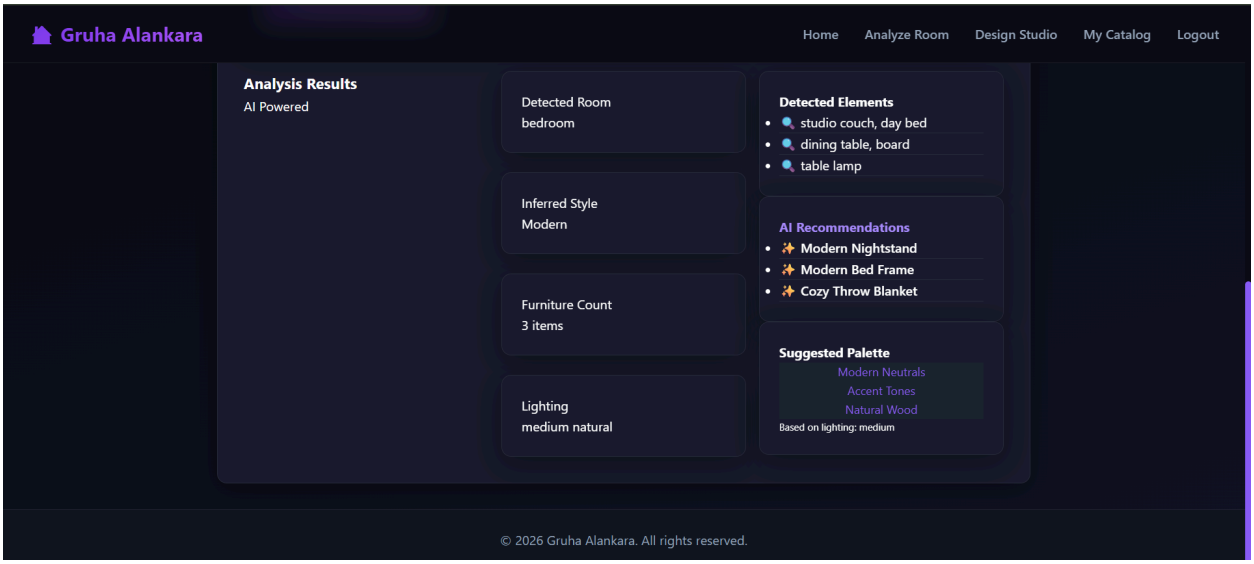
7.1 User Authentication Module



This module allows users to register and log in securely. Passwords are stored using hashing techniques to ensure security.

The login system uses session management to maintain authenticated user access to design features.

7.2 Room Analysis Module

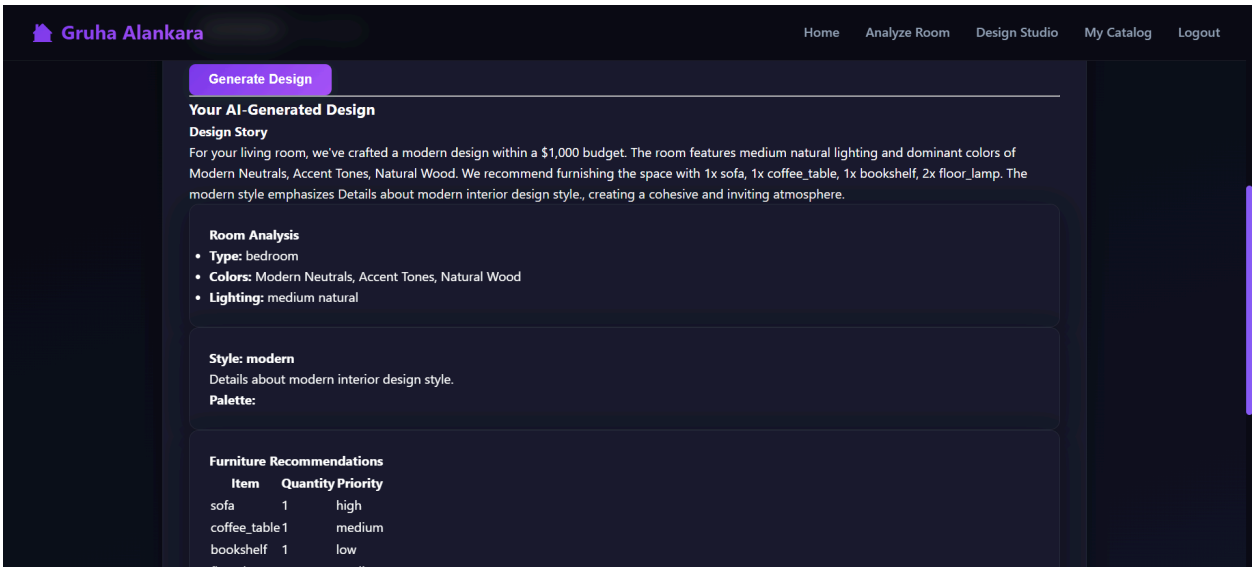


This module analyzes the uploaded room image and extracts useful information such as:

- room dimensions
- lighting conditions
- dominant color palette
- spatial constraints

The extracted information is used as input for the AI design recommendation system.

7.3 Style Recommendation Module



The system allows users to select preferred design styles such as:

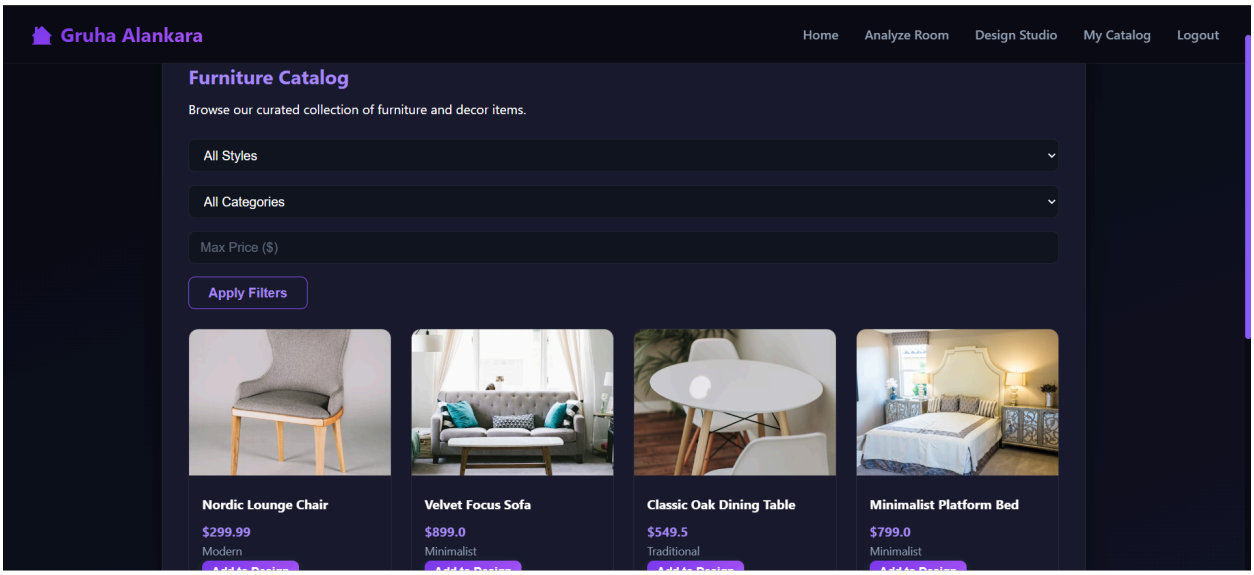
- Modern
- Minimalist
- Traditional
- Contemporary

Based on the selected style, the system generates design themes including color combinations and furniture styles.

7.4 Furniture Optimization Module

This module determines the optimal placement of furniture based on the room layout and available space. It ensures that recommended furniture fits within the room dimensions while maintaining aesthetic balance.

7.5 Furniture Catalog Module



The furniture catalog displays available furniture items categorized by type and price range. Users can browse through items recommended by the AI system.

7.6 Booking Module

The booking system allows users to select furniture items and simulate purchasing them. The system records the booking information and displays order details.

8. Implementation

The backend of the application is implemented using the Flask web framework. Flask routes handle different user interactions such as login, image upload, and design generation.

The AI pipeline coordinates multiple modules to analyze room images and produce design recommendations. The system also integrates browser-based camera access using JavaScript and the MediaDevices API.

The frontend is built using responsive HTML templates with a dark-themed CSS design to enhance user experience.

9. Results

The system successfully generates interior design recommendations based on uploaded room images and selected style preferences.

Users can explore different design layouts, furniture suggestions, and color schemes. The AI modules provide structured outputs including furniture recommendations and layout guidance.

The system demonstrates how AI can simplify the interior design process and provide personalized suggestions to users.

10. Conclusion

The Gruha Alankara project demonstrates the integration of artificial intelligence and web technologies to create an intelligent interior design platform.

The system allows users to analyze rooms, generate design recommendations, and explore furniture options through an intuitive interface. By combining AI models, image processing techniques, and a modern web framework, the project provides an innovative solution for automated interior design assistance.

Future improvements may include real-time AR visualization, advanced machine learning models for better room analysis, and integration with real-world furniture marketplaces.